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(1) Research Program/Project/Study Title: THE POTENTIAL OF CALAMANSI SEED AS MATERIAL FOR COFFEE-LIKE BEVERAGE	
Proponent Name:	REEVEN SIMON
Academic Rank:	Asst. prof 3
Co-Authors (with academic rank):	LEAH B. YAKIT– Asso. Prof 1 KRIS B. ROXAS – Asso. Prof . 1 CARLO DEGALA – Contractual Instructor
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Program or Project has component (if no proceed to 2)? ☐ Yes ☒ No

Component Title:	
Component Leader:	
Academic Rank:	

(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields ☐ Natural/BiologicalSciences ☒ Food Research/ ☐ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster		(3) Research Center [Check the concerned center] ☐ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☒ FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center	
(4) Site of implementation			
		Municipality/City:	Pontevedra
		Province:	Capiz
		Region:	6
(5) Classification Research: Development: ☐ Basic ☒ Pilot Testing ☒ Applied ☐ Tech. Promotion/Commercialization		(6) Mode of Implementation ☐ Single Authorship/Proponent ☒ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)	
(7) Sector/Commodity Calamansi		(8) Discipline/Field of Specialization Food Science and Technology	

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(9) Rationale (State the research problem and significance of the project to the current needs of the country. The proposal should justify resource expenditure. A typical justification would include a brief introduction, a general statement concerning the historical basis for R & D and, thrusts being addressed, utilization of the expected output and the impact information generated will have on science, the target users and the country.)

Calamansi (*Citrofortunella microcarpa*), is also known as *Laranghita*, *Calamondin*, Philippine Lime or Philippine Lemon. It is an indigenous crop and is predominantly cultivated in the Philippines. A total land area of about 19.40 thousand hectare is planted with calamansi from January to June 2023 as per recorded of the Philippine Statistics Authority (PSA, nd). Calamansi is considered as one of the high valued agricultural crops with economic importance (SEAMEO-SEARCA, 2019). It accounts for 20 percent of the total citrus exports of the country as cited by Lamberte (2018). In 2022, PSA reported that the country produced 118,248 metric tons (MT) of calamansi. The value of which amounted to approximately 3.86 billion Philippine pesos.

Calamansi is a versatile fruit known for its diverse use in the the Filipino household. It has a widespread use in local cuisine because of its unique flavor profile that is naturally sour with a hint of sweetness. Also it has the ability to enhance a myriad of dishes. Additionally, calamansi is commonly used to prepare various condiments, beverages, dishes, marinades, and preserves. Beyond its culinary uses, calamansi boasts a range of health benefits, containing high levels of vitamin C, antioxidants, and other essential nutrients. Calamansi fruit is also used for commercial processing of bottled concentrates, ready-to-drink juices, as well as a primary ingredient for cleaning and cosmetic products. Evidently, the there has been an increase in products derived from calamansi and its culinary and pharmaceutical application expanded.

The surge in demand for calamansi products reflects a growing appreciation for its unique flavor and diverse applications. However, this increase in popularity has brought about a parallel challenge, that is the escalating waste production. PSA estimated that a volume of 81,591 MT waste comes from the calamansi industry in the year 2022, composed mostly of its unused parts particularly its peel and seeds. This by products which are considered waste from the processing of calamansi, such as the seed, still have potential value which can be exploited to address the environmental and economic impact associated with its disposal.

This study, therefore, aims to explore the novelty of utilizing calamansi seed as an alternative material for the production of coffee-like beverage. It is anchored on sustainable development goal (SDG) 12: Responsible Consumption and Production with the purpose of finding sustainable ways to utilize and consume what our planet can provide and reduce the harmful impacts of waste generation.

(10) Objectives/Statement of the Problem (for Social Sciences) (Statement of the specific purpose to address the problem areas of the project.)

Objectives of the Study

In general, this study aims to investigate the feasibility and potential use of calamansi as a viable and sustainable alternative material in the production of coffee-like beverage.

Specifically, it aims to:

1. determine the the effect of fermentation time on the sensory characteristics of the beverage produced from roasted calamansi seed.
2. determine the effect of pre-processing methods on the taste characteristics of characterstics of calamansi seed beverage, focusing on the minimization of acrid and bitter flavor
3. optimize time and temperature of roasting calamansi seed for a desirable taste quality of the coffee-like beverage;
5. analyze the chemical composition of the developed beverage from calamansi seed.

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(11) Review of Literature/Prior Art Search (Related researchers/activities that have been conducted. State-of – the-art of current technology / information from which the project proposal will take-off).

Closest Prior Art:

WO2023028616A1 Alternative coffee beverages (Ganesh, et al., 2022)

Coffee is typically brewed from roasted and ground seed of coffea plant and is one of the most consumes products world wide, with daily consumption of about 2.25 billion cups. The high global demand for coffee imposes significant negative externalities on th environment and on coffee-producing countries. This results to the conversion of forest and farmlands to coffee plantations. In addition, coffee growth, processing, and transportation entail high water and carbon usage: 140 L of water are needed to produce 1 cup of coffee, and 1 kg of coffee creates 17 kg of CO₂ equivalents (Ganesh et al., 2022).

The study of Hassard, et al. (2014) looks into how different ways of making and packaging coffee affect the environment, emphasizing the need to carefully choose how we measure their carbon footprint. The research shows that this measurement can change a lot depending on what we compare it to. It finds that the main things causing emissions in the coffee industry are milk, the cans used for packaging, and the process of making green coffee beans. Interestingly, even though canned coffee makes up only about 17% of the coffee market, it is responsible for about half of the coffee-related carbon emissions in the country. The study also examines how effective current rewards are for customers who bring their own cups, compared to the impact of carbon taxes. It turns out that these rewards value reducing carbon far more than the usual prices in the carbon market.

Commercial coffee contains a variety of chemical substances, including caffeine, linoleic acid, trigonelline, chlorogenic acids, diterpenes like cafestol and kahweol, along with carbohydrates, proteins, peptides, amino acids, minerals, and lipids. These elements play a significant role in shaping the flavor, color, scent, and health impacts of coffee. Interestingly, some fruit seeds share several beneficial components with coffee, such as carbohydrates, linoleic acid, lipids, proteins, peptides, and amino acids. However, the lack of caffeine in okra seeds might enhance the efficacy of these health-promoting elements found in coffee (Buela, 2017).

(12) Scientific Basis/Theoretical Framework (Include concepts. theories and variables that indicate relevant relationships to be tested.

This study is based on the idea of using resources wisely and reducing waste, which are big challenges in today's world. This research focuses on the concept of a circular economy (Clark, et al., 2020), which means finding new uses for things that are usually thrown away, like calamansi seeds from the food industry. This framework emphasizes transforming end-of-life products into valuable resources, hence, minimize waste generation and transition from a consumption-focused model to one of sufficiency and resourcefulness. This approach helps in cutting down waste and making the most of what is available. The framework also advocates for a transition to a low carbon economy, underpinned by the use of renewable resources and green manufacturing practices. This includes the incorporation of innovative green technologies and bioprocessing methods. Such practices are instrumental in the production of clean energy, environmentally friendly chemicals, and safe, high-quality products with enhanced functionality.

The study also looks at using renewable resources and eco-friendly methods to create new products (Tsouko, et al., 2023). The goal is to make a coffee-like drink from calamansi seeds, using green technology. This method is not only good for the environment but also turns a waste product into something valuable. By doing this, a new kind of coffee that is sustainable and reduces the environmental impact compared to regular coffee can be materialized.

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(13) Methodology (Details such as; a) variables to be measured and evaluated; b) treatments to be used and their layout; c) experimental procedures and design; d) statistical analysis; e) evaluation method and observations to be made, strategies for implementation.)

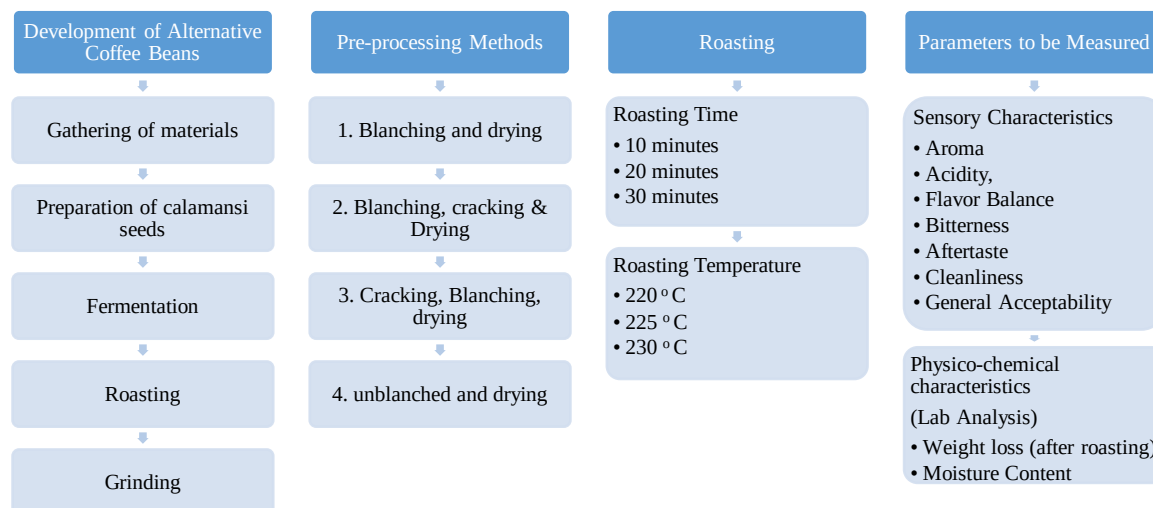


Figure 1. Summary of Experiment Protocol

(14) Major Activities/Work plan (Enumerate in chronological order the task to be undertaken. Use gantt chart.)

Activities	Month 1	Month 2	Month 3	Month 4	Month 5	Month 6
1. Gathering of materials						
2. Preparation of Calamansi Seeds to be Used as Alternative Coffee Beans						
3. Roasting of Alternative Coffee Beans						
4. Laboratory Analysis						
5. Sensory Evaluation						
6. Data Gathering						
7. Data Analysis						
8. Writing of Report						

(15) Expected Output (Indicate the specific products, processes or services which the project is expected to produce; quantify when possible)

Publication

- at least 1 original article on the product development and testing
- at least 1 technical manual/guide

Patent/Intellectual Property

- at least 1 IP (Patent/UM) applications – method of preparation

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Product

- at least 1 coffee drink formulation
- at least 2 processes - process of preparing the coffee beans and process of roasting

People Service

- at least two (2) young faculty and research personnel trained
- (1) Invention disclosure report
- (1) prior art search report

(15) Technology Transfer Activity Design (Indicate the specific products, activity design, processes or services to be transferred as extension activities)

Training on the processing methods of producing low-sugar calamansi juice

(16) Targets Beneficiaries (Who the clientele are and what are the expected outcome/effects of the use of the projects output)

Farmers, food processors, health enthusiasts, consumers, entrepreneurs, researchers

(17) Personnel Requirement

Name	Role (Program, Project, Study Leader)	Field of Specialization
REEVEN SIMON	Project Leader	Food Service Management
LEAH B. YAKIT	Project Staff	Food Science
KRIS B. ROXAS	Project Staff	Food Chemistry
CARLO DEGALA	Project Staff	Food Service Management

Percent time devoted to the Program/ Project/ Study

Program/Project Leader (in hours): 15 hrs/wk

Program/Project Members (in hours):

(18) Budgetary Requirement

Line Item Budget							
Particulars	Month						Total
	1	2	3	4	5	6	
I. PERSONAL SERVICES							
A. SALARIES AND WAGES							
Sub-total for PS							-
II. MAINTENANCE AND OTHER OPERATING EXPENSES							
Travelling Expenses		640.00	640.00	640.00	320.00	320.00	2,560.00
Office supplies	1,000.00						1,000.00
Ingredients		5,000.00					5,000.00
Other Professional Expenses (Lab Analysis)				10,000.00			10,000.00
Sub-total for MOOE	1,000.00	5,640.00	640.00	10,640.00	320.00	320.00	18,560.00
III. EQUIPMENT OUTLAY							
High Power Coffee Grinder		5,000.00					
Sub-total for EO		5,000.00					5,000.00
Grand Total	1,000.00	10,640.00	640.00	10,640.00	320.00	320.00	23,560.00

(19) Literature Cited (include only those references published in reputable journals, and at least that from CAPSU Research Journal, <http://researchjournal.capsu.edu.ph/>)

Buela, A. (2019). Okra Seeds: Alternative Coffee. Ascendens Asia Journal of Multidisciplinary Research Abstracts. 3 (20)

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Ganesh, V, Gonzalez, MJ, Molina, A, Saenz, M., Shumaker, A., Youngblut, M., Hadevan, S. (2022). Alternative Coffee Beverages. WO2023028616A1. World Intellectual Property Organization

Lamberte, MB. (2018). Smallholder Commodity Systems in High-Value Crops: The Case of Calamansi and Jackfruit in the Philippines in <https://serp-p.ids.gov.ph/publication/public/view?slug=smallholder-commodity-systems-in-high-value-crops-the-case-of-calamansi-and-jackfruit-in-the-philip> accessed on January 24, 2024.

PSA (nd). Major Fruit Crops Quarterly Bulletin, April-June 2023. Accessed from <https://psa.gov.ph/major-fruit-crops/calamansi> on January 25, 2024.

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Tsouko, E, Maina, S, Alexandri, M, Papapostolou, H, and Koutinas, A. (2023). Recovery of Value-Added Products and Biological Conversion of Coffee and Citrus Processing Waste using Green Technologies. Retrieved on: January 30, 2024 from: <https://books.google.com.ph/books?id=FACoEAAAQBAJ&lpg=PA111&ots=6mZXAb6WKD&dq=coffee%20from%20citrus%20seeds&lr&pg=PA111#v=onepage&q&f=false>

(20) Capsule Curriculum Vitae (One-page brief C.V. only)